

# Yeaseen Arafat

Salt Lake City, Utah, USA

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## RESEARCH SUMMARY

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My research focuses on program analysis and software testing, with an emphasis on improving the correctness of cross-language program transformations and software reliability.

## EDUCATION

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### 🎓 The University of Utah

*Ph.D. in Computer Science*

*Aug 2023 – Apr 2028 (Expected)*

*M.S. in Computer Science*

*Aug 2023 – Dec 2025*

### 🎓 Bangladesh University of Engineering and Technology (BUET)

*B.Sc. Engg. in Computer Science and Engineering*

*Feb 2015 – Apr 2019*

## PROFESSIONAL EXPERIENCE

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### Research Intern

*Aug 2026 – Dec 2026*

*Aarno Labs*

*Remote*

- Developing a symbolic-execution-based tool for generating panic-triggering inputs in translated Rust programs.

### Graduate Research Assistant

*May 2024 – Present*

*FuTURES<sup>3</sup> Lab, The University of Utah.*

*Utah, USA*

- Developed a symbolic-execution-based translation validation technique for cross-language software migrations.
- Built PROGnosticator, an LLM-assisted construct-oriented transpiler fuzzer that discovered **64** new bugs—all confirmed by developers—across seven C, Go, and JavaScript transpilers.
- Designed TeTRIS, a general-purpose mutational fuzzing framework that discovered **12** previously unknown transpiler bugs, all confirmed by developers, across seven C, Go, and Haxe transpilers.
- Implemented a cross-language parser that normalizes C, Go, and Haxe into a neutral Abstract Syntax Tree with scope and type inference, enabling syntax- and semantics-preserving uniform program mutations.

### Lecturer

*June 2019 – July 2023*

*Department of Computer Science and Engineering, Southeast University*

*Dhaka, Bangladesh*

- Taught core computer science courses and supervised undergraduate theses in machine learning and data analysis, contributing to curriculum design and program modernization.

### Undergraduate Student Researcher

*Apr 2018 – Apr 2019*

*Bangladesh University of Engineering and Technology*

- Designed a machine learning-based intrusion detection system for the WiFi MAC layer (IEEE 802.11), implemented in Network Simulator-2, using statistical network traffic features to identify malicious behavior.

## PUBLICATIONS

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- **Yeaseen Arafat**, Stefan Nagy, “PROGnosticator: Testing Source-to-Source Code Translators via Construct-oriented Fuzzing”, *ACM International Conference on Foundations of Software Engineering (FSE 2026)*. [doi] [pdf] [code]
- **Yeaseen Arafat**, Stefan Nagy, “TeTRIS: General-purpose Fuzzing for Translation Bugs in Source-to-Source Code Transpilers”, *Annual Computer Security Applications Conference (ACSAC 2025)*. [doi] [pdf] [code]
- **Yeaseen Arafat**, Kazi Samin Yeaser, Ashikur Rahman, Arnab Dasgupta, “A Machine Learning based Approach for Protecting Wireless Networks Against DoS Attacks”, *International Conference on Networking, Systems and Security (NSysS 2020)*. [doi] [pdf] [code]

## OPEN SOURCE CONTRIBUTIONS

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- **Bug Discovery:** Discovered **90+** developer-confirmed previously-unknown correctness bugs across projects such as `c2rust`, `Go compiler`, `Zig`, `cxgo`, `swc`, `go2hx` and `tmux-rs`. [[FuTURES<sup>3</sup> Lab's Bug Tracker](#)]
- **Upstream Contributions:** Merged PRs to `Node.js`, `LLVM`, and `C2Rust` on filesystem correctness, test determinism, and language semantics, with additional fixes in `CxGo` and `Polyglot`. [[details](#)]
- **Security Disclosures:** Discovered two published vulnerabilities in `libyang` [[1](#)] [[2](#)], a confirmed vulnerability in `libarchive` [[3](#)], and a critical key-exposure vulnerability in a mobile banking app [[details](#)].
- **TeTRIS:** Publicly reviewed and discussed by the *Go2Hx* developer community. [[details](#)]

## TALKS AND PRESENTATIONS

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- **FSE 2026** — PROGnosticator [[slides](#)]
- **ACSAC 2025** — TeTRIS [[slides](#)]
- **NSysS 2020** — Wireless Network DoS Detection [[talk](#)] [[slides](#)]

## TECHNICAL PROJECTS

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- KerFuzz: Linux Kernel Fuzzer** | *C, Bash, GDB, Syzkaller, QEMU*  [GitHub](#)
- Designed a syscall-level kernel fuzzing workflow using Syzkaller to discover bugs in the Linux kernel.
  - Instrumented and analyzed faulting executions using GDB and custom Bash scripts for postmortem triage.
- Cyber-Physical Systems Security CTFs** | *OpenPLC, Modbus/TCP, AWS DeepRacer*  [GitHub](#)
- **CTF 1:** Detected PLC attacks using code-integrity checks and Modbus/TCP runtime monitoring.
  - **CTF 2:** Tested AWS DeepRacer perception using FGSM adversarial examples and custom stop-sign detection.
- CyberShield-Spectrum** | *Virtualized Security Operations Lab*  [GitHub](#)
- Built and secured a virtual enterprise network with VPN, firewall, IDS, jump, backup, and vulnerable web servers.
  - Conducted penetration testing, reconnaissance, web exploitation, and network, memory, and disk threat hunting.

## AWARDS & SCHOLARSHIP

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- **Graduate Research Fellowship**, University of Utah (Aug–Dec 2023)
- **University Merit Scholarship**, BUET (2015–2019)
- **Junior Government Scholarship**, Chittagong Education Board, Bangladesh (2009)

## OUTREACH ACTIVITIES

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- **Artifact Evaluation Committee Member:** [ACM CCS 2024](#), ACSAC 2025
- **Graduate Teaching Assistant:** CSE-4480 (Computer Networks), University of Utah (Spring 2024). Supported 170+ students via office hours, grading, and final evaluations.
- **Organizing Committee Member:** ACM ICPC ASIA Regional, Dhaka Site 2019

## TECHNICAL SKILLS

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**Programming Languages:** C/C++, Rust, LLVM IR, Go, Python, Java, Assembly (x86, MIPS), Wasm  
**Security & Fuzzing:** AFL++, KLEE, Kani, Syzkaller, QEMU, Burp Suite, Wireshark, sqlmap, Snort, Pwndbg  
**Systems:** LLVM, Flex/Bison/YACC, Nachos, Make, Static Analysis  
**Machine Learning:** TensorFlow, PyTorch, scikit-learn, Pandas, NumPy  
**Web & DevOps:** React, Node.js, Express, Postman, Docker, GitHub Actions, GraphQL, REST API